The Haber Process

Exam code: 0620 & 0971

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The Haber process

Extended tier only

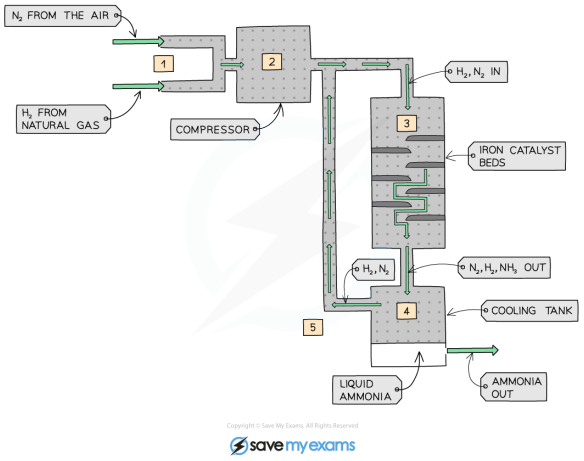
- Ammonia is manufactured in an **exothermic** reaction called the **Haber process** which occurs in five stages:
- **Stage 1**
  - H_2 is obtained from methane
  - N_2 is obtained from the air
  - They are pumped into the compressor through a pipe
- **Stage 2**
  - Inside the compressor, the gases are compressed to around **20 000 kPa** or **200 atmospheres**
- **Stage 3**
  - The pressurised gases are pumped into a tank containing layers of an **iron catalyst** at a temperature of **450 °C**
  - Some of the hydrogen and nitrogen react to form ammonia:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

- **Stage 4**
  - Unreacted H_2 and N_2 and the ammonia product pass into a cooling tank
  - The ammonia is liquefied and removed to pressurised storage vessels

Help me understand

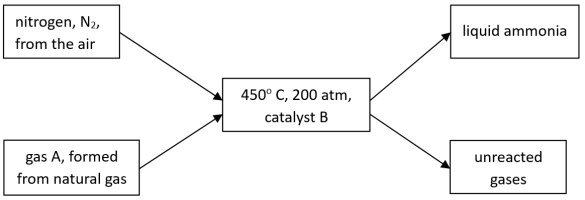
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



The production of ammonia by the Haber process

Worked Example

Ammonia is produced during the Haber Process. The reaction is summarised in the diagram below.



1. Name gas **A**.
2. Name catalyst **B** used and state why it is used

Answer:

1. Gas **A** is hydrogen / H_2.
2. Catalyst **B** is iron, which is used to speed up the reaction / increase the rate of reaction.

Examiner Tips and Tricks

Examiners often comment that while many students can state the overall reaction, they frequently lose marks by not knowing the specific details.

To secure top marks, make sure you can recall:

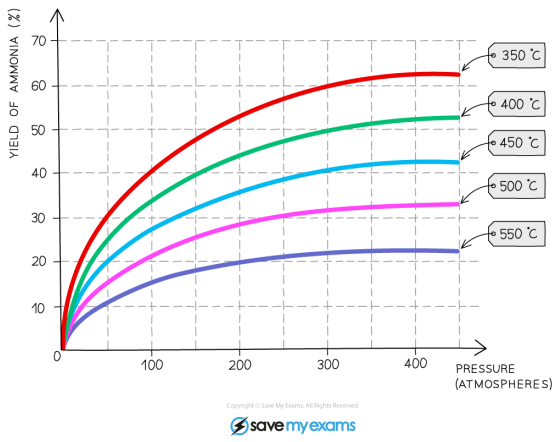
- The **sources of the raw materials**:
  - **Nitrogen** from the fractional distillation of liquid **air**
  - **Hydrogen** from **natural gas** (methane)

- Do not confuse it with catalysts from other industrial processes, like 'vanadium oxide' and 'enzymes'

Explaining the conditions in the Haber process

Extended tier only

- Chemists' knowledge of the energy changes and factors affecting reaction rates can be used to predict the best possible conditions to make the most ammonia in the fastest possible time
- However, sometimes those conditions are contradictory and choices have to be made between factors that improve the yield of ammonia and those that speed up the reaction
- The graph below illustrates the effects of changing temperature and pressure on the yield of ammonia obtained



The yield of ammonia produced changes with changes made to temperature and pressure

- From the graph:
  - As the **pressure increases,** the percentage yield increases
    - This is shown by following any of the curved lines
  - As the temperature **decreases,** the percentage yield increases
    - This is shown by following any vertical line upwards from the x-axisthat
- The actual conditions used must be chosen depending on a number of economical, chemical and practical considerations

Economic considerations

- Like all industries, companies that manufacture and sell chemical goods do so to make a profit

- The availability and cost of **raw materials** is a major consideration which must be studied well before any decisions are taken
- In the Haber Process the raw materials are readily available and inexpensive to purify:
  - **Nitrogen** - from the **air**
  - **Hydrogen**- from **natural gas**

- If the cost of extraction of raw materials is too high or they are unavailable then the process is no longer **economically viable**
- Many industrial processes require huge amounts of **heat** and **pressure** which is very expensive to maintain
- Production energy costs are also a factor to be considered carefully and alongside the raw materials issue

Temperature: 450°C

- **High** temperature favours the reverse reaction as it is endothermic
  - So, a higher yield of **reactants** will be made

- **Low** temperature favours the forward reaction as it is exothermic
  - So, a higher yield of **products** will be made

- However, at low temperature the rate of reaction is very **slow**
- So, 450 °C is a compromise temperature between having a **lower yield** of products but being made more quickly

Pressure: 200 atm

- **Low** pressure favours the reverse reaction as there are more moles of gaseous reactant
  - So, a higher yield of **reactants** will be made

- **High** pressure favours the forward reaction as there are **fewer** moles of gaseous product
  - So, a higher yield of **products** will be made

- However, high pressure can be dangerous and very expensive equipment is needed
- So, 200 atmospheres is a **compromise** pressure between a lower yield of products being made **safely** and **economically**

Catalyst: Iron

- The presence of a catalyst does **not** affect the position of equilibrium but it does increase the **rate** at which equilibrium is reached
- This is because the catalyst increases the rate of **both** the forward and backward reactions by the same amount (by providing an alternative pathway requiring **lower activation energy**)
- As a result, the **concentration** of reactants and products is nevertheless the **same** at equilibrium as it would be without the catalyst.
  So a catalyst is used as it helps the reaction reach equilibrium quicker

even **higher temperature**, increasing **costs** and decreasing **yield** as the higher temperature **decomposes** more of the NH_3 molecules

Examiner Tips and Tricks

The reaction conditions chosen for the Haber process are not ideal in terms of the yield but do provide balance between product yield, reaction rate and production cost.

These are called **compromise conditions** as they are chosen to give a good compromise between the yield, rate and cost.

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